# Echolocation

Ghost bats live in northern Australia and are Australia's only carnivorous bat species. Ghost bats have long, large ears that are very sensitive to sound. They use **echolocation** to hunt at night for large insects, lizards, birds, frogs and small bats. Used by many bat species, echolocation involves using sound waves to locate food and find their way around in the dark.

The sound waves bounce back from animals and objects around the bats. They listen for the returning waves to work out the size, location and direction of movement of animals nearby. If the waves return quickly, they know the animal is close by. If the sound waves are intense, this tells them that the animal is large. If the waves return first to the bat's left ear, then moments later to its right ear, this means the animal is on their left.

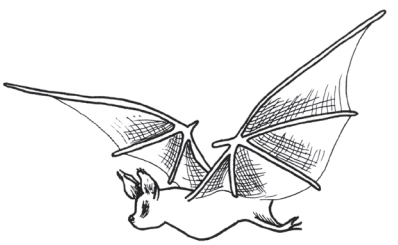
☞ **1** Describe a ghost bat's ear. _____

**2** Why do ghost bats use echolocation? _____

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**3** Draw sound waves coming from the ghost bat in blue and bouncing back from the bat's prey in red.





**4** Where would the prey be if the sound waves return first to the bat's right ear?

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**5** If the echoing sound wave is faint, what size would the animal be? _____

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**6** If the sound waves take a while to return, how far away is the prey? _____

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